

Puneeth Muthineni

AI Engineer · LLMs · RAG · Multi-Agent Systems · Python · AWS · CI/CD

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SUMMARY

AI Engineer and MS Computer Science graduate (UMass Boston, 2026) with 1+ year of professional Python experience designing, building, and deploying production AI systems built on LLMs, RAG, and multi-agent workflows. Architects LLM orchestration and tool-using agent pipelines with LangGraph and LangChain, integrating model APIs and SDKs (OpenAI, Gemini, Claude) with REST services, databases, and retrieval layers. Builds and maintains scalable data ingestion and retrieval pipelines (FAISS vector search, semantic retrieval) and evaluation frameworks measuring agent correctness, latency, and reliability. Ships production-grade, observable services on AWS (S3, EC2, Lambda) with CI/CD and monitoring, applying REST best practices and structured validation. Strong written and verbal communication; published NLP researcher and AWS-certified Machine Learning Engineer.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, SQL, Go, Bash

Frontend / Web: React, JavaScript, HTML, CSS, REST APIs

Backend: FastAPI (primary), Flask, Node.js / Express, Spring Boot, Django REST Framework, Pydantic

Databases: PostgreSQL, MongoDB (SQL & NoSQL), SQLite, Oracle, DB2; vector databases (FAISS) — schema design, normalization, data modeling

DevOps & CI/CD: Docker, Docker Compose, Kubernetes, GitHub Actions, Jenkins, pytest, JUnit; testing, monitoring & operational ownership

Cloud & Tools: AWS (S3, EC2, Lambda), Azure, GCP, Git / GitHub; Linux/Unix development environments; AI-assisted development with GitHub Copilot & Gemini

LLM & Agent Frameworks: LangGraph, LangChain, OpenAI Agents SDK; LLM APIs (OpenAI, Gemini, Claude); RAG & FAISS vector search; agent orchestration, prompt engineering, structured outputs; HuggingFace Transformers, BERT

ML & Data: scikit-learn, XGBoost, pandas, NumPy, MLflow, Optuna, PyTorch

WORK EXPERIENCE

Freelance Software Developer · Acentle · Remote Jan 2025 – Apr 2026

- Architected and shipped SyncDev Spatial, a real-time, networked client-server backend service (Python + WebSockets) exposing 10+ REST API endpoints with LLM-powered features, containerized and deployed end to end via Docker; reduced manual operational overhead by ~40%.
- Built and operated a GitHub Actions CI/CD pipeline automating build, lint, and test on every push, cutting deployment-validation time ~30% across UAT and production, with monitoring for reliable releases.
- Collaborated cross-functionally with product owners, developers, and QA in an Agile model; used GitHub Copilot and Gemini to accelerate delivery and improve code quality across 5+ production releases.

Software Developer Intern · Nice Interactive · Remote Jan 2024 – Jun 2024

- Developed and maintained backend features for web applications, gaining hands-on full-stack experience within structured SDLC practices.
- Wrote and executed SQL across 3+ application modules for data retrieval and reporting; contributed to Git workflows and CI/CD runs (branches, pull requests, code reviews) under senior-developer guidance.

PROJECTS

Unisphere AI Adaptive Learning Platform — Production RAG + Agents

Python · FastAPI · LangChain · LangGraph · FAISS · AWS

- Implemented a production RAG system on AWS (S3 + EC2) to improve LLM performance — document ingestion, FAISS vector database indexing, semantic retrieval, and validated data flow between the application and LLM inference layer.
- Built LangGraph agent frameworks applying Agentic AI principles for autonomous decision-making and task execution; shipped two production AI features with evaluation criteria for correctness, latency, and consistency.

CareCompass — Full-Stack GenAI Health Navigator

React · Node.js/Express · FastAPI · MongoDB · Docker Compose · GitHub Actions

- Built a full-stack application — React frontend, Node.js/Express gateway, and Python/FastAPI service — containerized with Docker Compose and deployed across environments.
- Engineered the API gateway (MongoDB persistence, structured validation, REST APIs) and a multi-step LangGraph agent workflow with structured outputs and a deterministic safety gate; CI via GitHub Actions (lint + pytest).

MLOps Churn Prediction — End-to-End ML Pipeline

Python · FastAPI · PostgreSQL · MLflow · Optuna · pytest · GitHub Actions

- Built a production FastAPI inference service backed by PostgreSQL with Pydantic validation, returning structured predictions with probability scores and risk levels.
- Designed a 14-test pytest evaluation framework and automated the ML lifecycle across 96 MLflow experiments (Optuna tuning) — improving XGBoost F1 by 3.3% and AUC by 2.9%, with CI on every commit.

EDUCATION

MS, Computer Science · University of Massachusetts Boston

Sept 2024 – May 2026

B.Tech, Computer Science & Engineering · VIT-AP University

Sept 2020 – June 2024

PUBLICATIONS & CERTIFICATIONS

- BERT-QA: Empowering Intelligent Question Answering with NLP and Entity Recognition — published 2024.
- AWS Certified Machine Learning Engineer – Associate · Google AI Essentials.